Craftax runs compared: achievements and what unlocks them

21 runs, 1.1M env steps each, size50m, one training seed per run. Every run scored on the same 30 fixed worlds (episode i built from seed i; one world hash across all runs) with tools/death_eval.py. Updated 2026-10-14.

Every number is the mean episode-final achievement count over 30 evaluation episodes, ± its standard error over those worlds; unlock rates are the share of episodes in which an achievement fired; return is the mean raw Craftax return as a % of the maximum (226). An outside audit (2026-10-11, docs/audit-2026-10-11.md) found that earlier evaluations only shared the first world: later worlds depended on how long earlier episodes ran. All checkpoints were re-scored on fixed worlds, so every comparison below is paired world by world. The error bars cover the worlds only, not training seeds -- every run is one seed.

The same audit found three RSSM-2 bugs (non-moves counted as a step down, a new episode inheriting the old one's state, and training seeing up to 7 future steps). Honest map, B2, v1.2 and v2 were retrained with every fix (bold rows). None improved: B2 8.27 vs 8.40, v2 7.17 vs 6.93, honest map 5.70 vs 6.17 -- all within noise -- and v1.2 fell from 9.97 to 8.77 (paired -1.20 ± 0.57), which one seed cannot tell apart from seed variation.

What holds up on the fixed code: masking impossible actions (B2 vs honest map +2.57 ± 0.48, t = 5.3); v2 over vanilla DreamerV3 (+2.53 ± 0.44, t = 5.8); the honest map over vanilla (+1.07 ± 0.24), though that run also has the potential shaping. What is not established: the manager over B2 (v1.2 fix vs B2 fix +0.50 ± 0.57), the order among v1.1/v1.2/v1.3, and curiosity (v2 vs v2-cur +0.40 ± 0.39). Two more training seeds of the fixed B2, v1.2 and v2 are running.

Normalized return, raw Craftax reward on the fixed worlds: best 4.04% (v1.2, old code), 3.48% for v1.2 fixed, 3.32% for B2, 2.77% for v2 fixed, 1.65% for vanilla DreamerV3, against 7.09% for ITC (transformer world model, pixels). The B2 and v1.x runs read the game's action rules and hand-written goals; v2 keeps only the action rules.

Every run, side by side

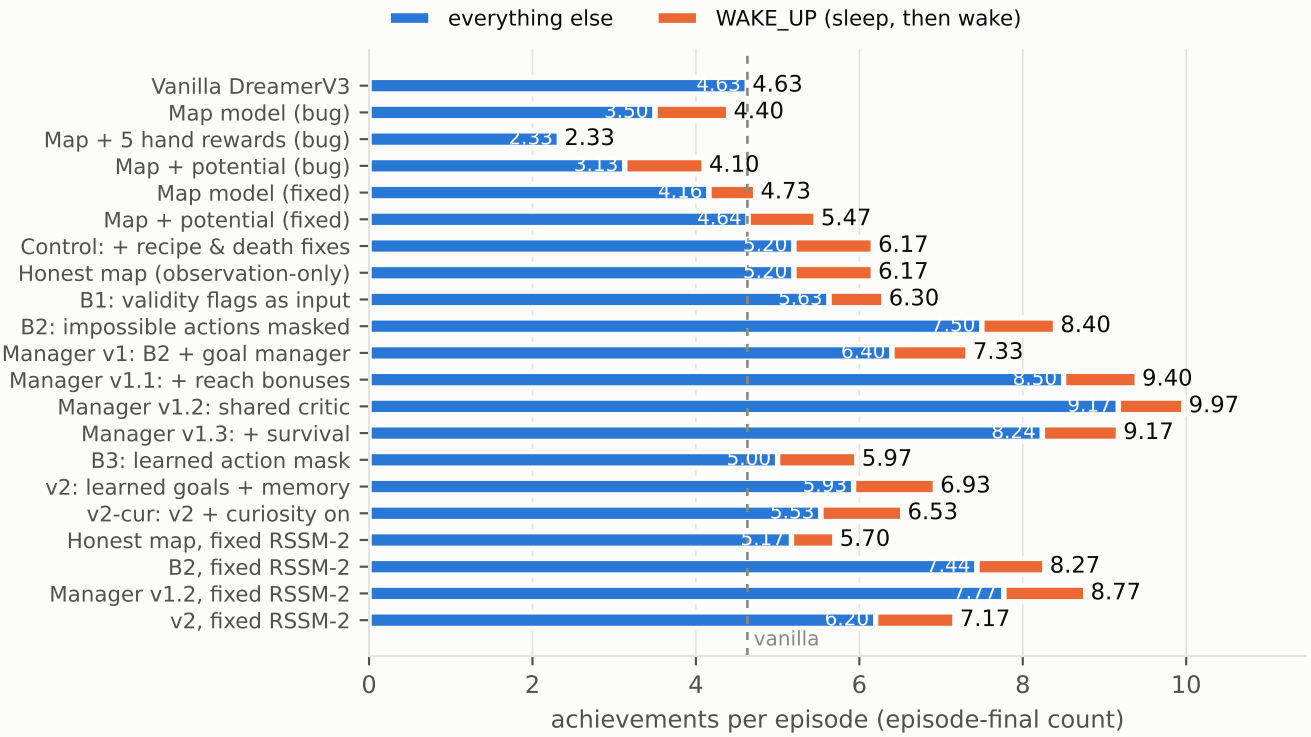
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run	ach. ± SE	w/o wake	life	return	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	4.63 ± 0.22	4.63	295	1.65%	63%	23%	3%	0%	77%
Map model (bug)	4.40 ± 0.29	3.50	263	1.55%	40%	0%	0%	0%	47%
Map + 5 hand rewards (bug)	2.33 ± 0.19	2.33	265	0.63%	0%	0%	0%	0%	43%
Map + potential (bug)	4.10 ± 0.19	3.13	275	1.42%	17%	0%	0%	0%	43%
Map model (fixed)	4.73 ± 0.29	4.16	276	1.70%	67%	10%	0%	0%	77%
Map + potential (fixed)	5.47 ± 0.34	4.64	264	2.02%	70%	33%	13%	17%	60%
Control: + recipe & death fixes	6.17 ± 0.23	5.20	304	2.33%	97%	0%	50%	0%	80%
Honest map (observation-only)	6.17 ± 0.34	5.20	280	2.33%	73%	40%	17%	7%	77%
B1: validity flags as input	6.30 ± 0.33	5.63	295	2.39%	93%	53%	7%	27%	77%
B2: impossible actions masked	8.40 ± 0.54	7.50	282	3.32%	93%	73%	37%	60%	47%
Manager v1: B2 + goal manager	7.33 ± 0.53	6.40	256	2.85%	90%	57%	27%	37%	67%
Manager v1.1: + reach bonuses	9.40 ± 0.54	8.50	274	3.76%	93%	80%	43%	80%	90%
Manager v1.2: shared critic	9.97 ± 0.36	9.17	262	4.04%	100%	93%	50%	87%	67%
Manager v1.3: + survival	9.17 ± 0.49	8.24	294	3.66%	97%	80%	43%	67%	97%
B3: learned action mask	5.97 ± 0.28	5.00	269	2.24%	73%	33%	10%	7%	77%
v2: learned goals + memory	6.93 ± 0.36	5.93	270	2.67%	93%	37%	20%	27%	73%
v2-cur: v2 + curiosity on	6.53 ± 0.22	5.53	319	2.50%	100%	23%	30%	7%	80%
Honest map, fixed RSSM-2	**5.70 ± 0.21**	**5.17**	**268**	**2.12%**	**87%**	**7%**	**50%**	**0%**	**77%**
B2, fixed RSSM-2	**8.27 ± 0.44**	**7.44**	**274**	**3.32%**	**97%**	**80%**	**40%**	**57%**	**67%**
Manager v1.2, fixed RSSM-2	**8.77 ± 0.52**	**7.77**	**319**	**3.48%**	**97%**	**67%**	**33%**	**67%**	**73%**
v2, fixed RSSM-2	**7.17 ± 0.44**	**6.20**	**276**	**2.77%**	**90%**	**53%**	**23%**	**30%**	**87%**

ach. ± SE = mean episode-final achievements ± standard error over the 30 worlds (one training seed, so seed variation is not included). w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. return = raw Craftax return, % of 226. Rates are % of episodes. Bold rows were trained on the code with the RSSM-2 and shaping fixes of 2026-10-11; the others are the original checkpoints, re-scored. 'bug' = imag_shift True (fixed 2026-09-20). B1/B2: validity flags as an encoder input / impossible actions masked. mgr v1-v1.3: B2 plus the two-level agent (docs/design-manager.md). B3: the mask learned instead of read from the rules. v2: learned goals, curiosity, RSSM-2 as a memory, rule mask kept (docs/design-v2.md); v2-cur: v2 with curiosity actually scaled.

Sleep inflates the map runs; the manager is real progress

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs

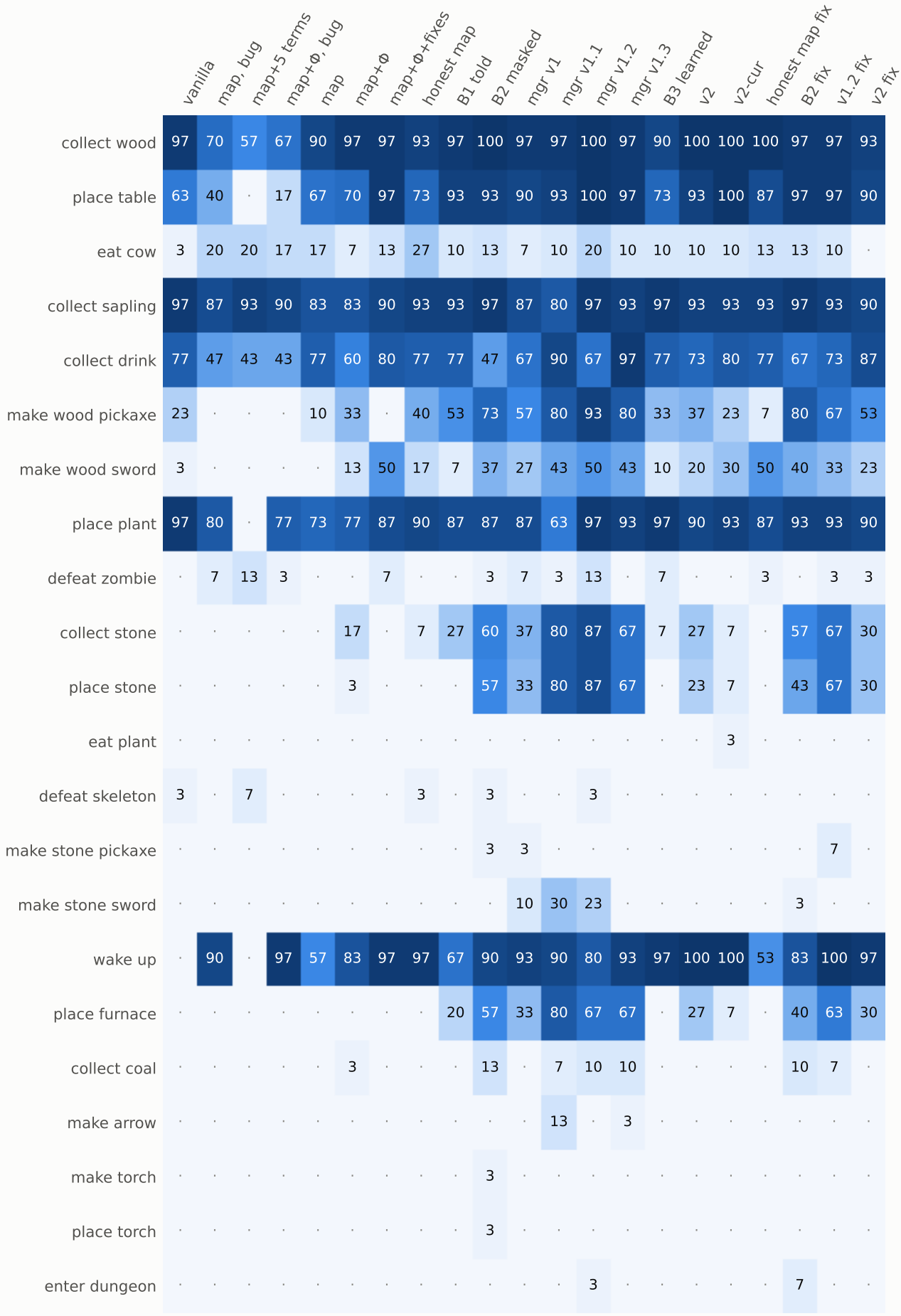


Unlock rate (% of episodes), early tech tree in dependency order

	collect wood	place table	make wood pickaxe	make wood sword	collect stone	place stone	place furnace	collect coal	make stone pickaxe
Vanilla DreamerV3	97	63	23	3	0	0	0	0	0
Map model (bug)	70	40	0	0	0	0	0	0	0
Map + 5 hand rewards (bug)	57	0	0	0	0	0	0	0	0
Map + potential (bug)	67	17	0	0	0	0	0	0	0
Map model (fixed)	90	67	10	0	0	0	0	0	0
Map + potential (fixed)	97	70	33	13	17	3	0	3	0
Control: + recipe & death fixes	97	97	0	50	0	0	0	0	0
Honest map (observation-only)	93	73	40	17	7	0	0	0	0
B1: validity flags as input	97	93	53	7	27	0	20	0	0
B2: impossible actions masked	100	93	73	37	60	57	57	13	3
Manager v1: B2 + goal manager	97	90	57	27	37	33	33	0	3
Manager v1.1: + reach bonuses	97	93	80	43	80	80	80	7	0
Manager v1.2: shared critic	100	100	93	50	87	87	67	10	0
Manager v1.3: + survival	97	97	80	43	67	67	67	10	0
B3: learned action mask	90	73	33	10	7	0	0	0	0
v2: learned goals + memory	100	93	37	20	27	23	27	0	0
v2-cur: v2 + curiosity on	100	100	23	30	7	7	7	0	0
Honest map, fixed RSSM-2	100	87	7	50	0	0	0	0	0
B2, fixed RSSM-2	97	97	80	40	57	43	40	10	0
Manager v1.2, fixed RSSM-2	97	97	67	33	67	67	63	7	7
v2, fixed RSSM-2	93	90	53	23	30	30	30	0	0

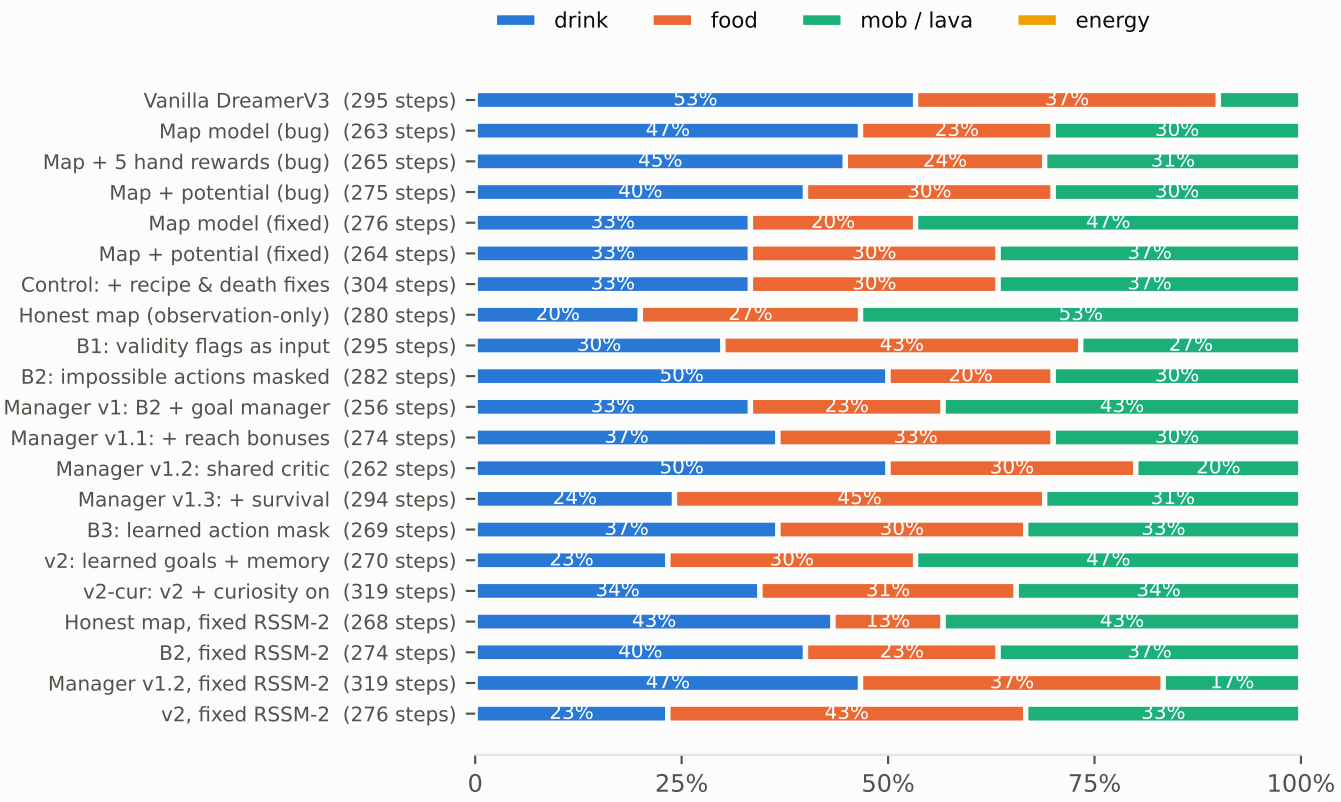
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 45 of 67 were never unlocked by any run.



Survival never moved

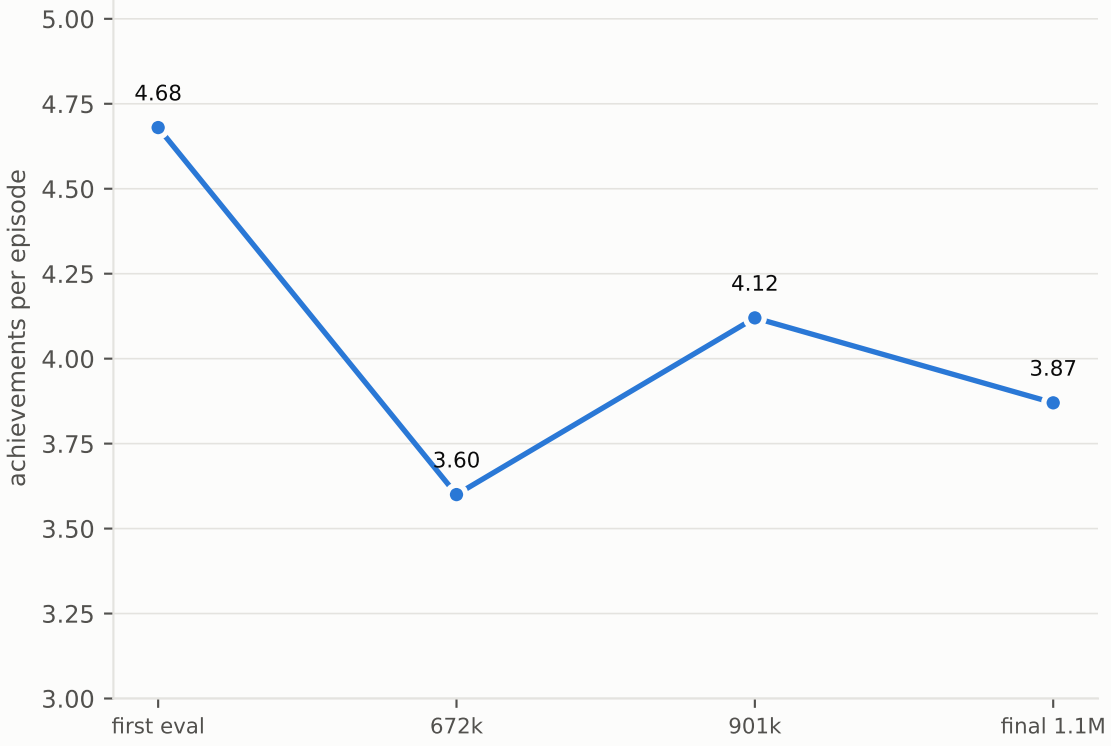
Cause of death (share of evaluation episodes) and lifespan; and how much one run moves between its own checkpoints



Random policy lives 261-284 steps; doing nothing lives 333. Map + potential (bug) has no recorded causes.

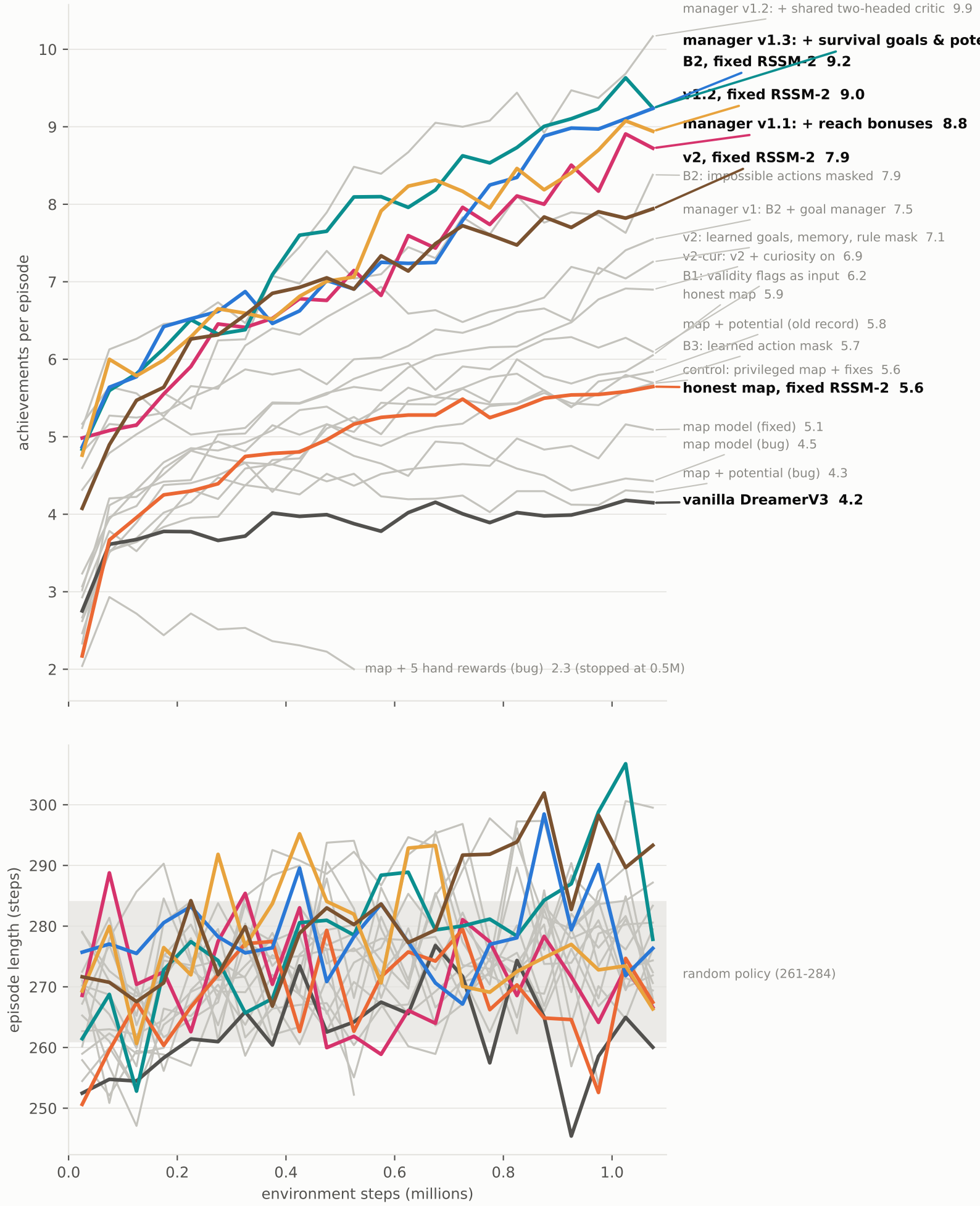
One run, four checkpoints: Map + potential (bug)

The same training run, scored at four points (25-30 episodes each), moves by 1.08 achievements. That is the scale of noise any comparison between single runs has to beat.



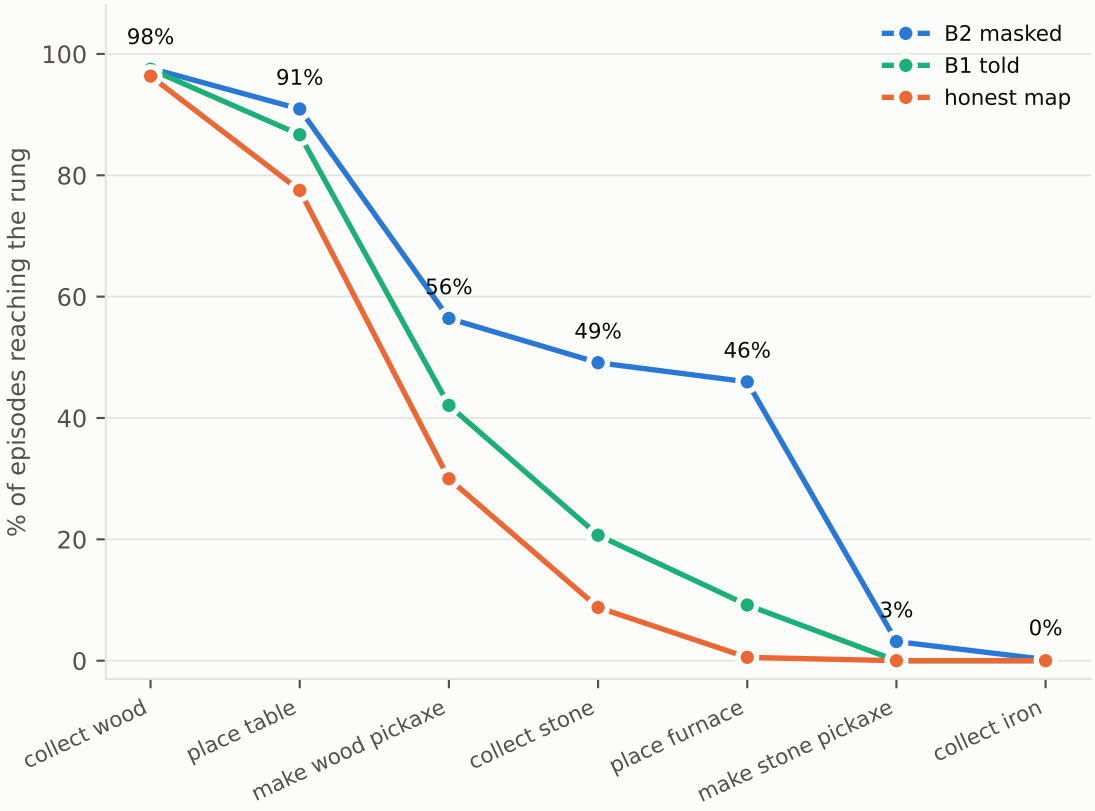
Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps; one seed per run.



Where B2 falls short

Last 150k training steps of each run (~500 episodes), decoded from replay by tools/diagnose_run.py



1. The stone pickaxe is the wall. B2 reaches the furnace in 46% of episodes but makes the stone pickaxe in 3%. Of the 249 episodes that collected stone, 73% held wood and stone at the same time, but only 22% ever stood next to a table while holding both, and 6% made the pickaxe. Two gaps: it rarely brings the ingredients back to a table (logistics), and when it does the chance lasts one step -- the single log usually goes into the wood sword or it walks off (measured in docs/b2-casebook.pdf).
2. Table to wood pickaxe leaks. 38% of the episodes that place a table never make the wood pickaxe. At a table holding wood it crafts every time, but every one of these episodes placed the table with its last two logs, and later logs go into another table rather than back to the first (docs/b2-casebook.pdf).
3. Survival. Training-window lifespan 285 (median 275) against 261-284 for a random policy. Thirst is still the first meter to run out in 45% of deaths and hunger in 24%: Craftax pays for the first drink only, so topping up is never rewarded. A hostile mob is adjacent at 61% of deaths; it dies in lava in 4% (the others: under 1%), the price of exploring further. Deaths in its sleep are rare (5%).
4. Wasted presses. 43% of presses still do nothing (experiment A): DO with nothing in front (26%), NOOP and walking into walls. Masking only removes the special actions; these basic ones are never masked.

What the results say, and what would settle the rest

0. Masking impossible actions is the one large, robust gain

B2 beats the same model without the mask by +2.57 ± 0.48 achievements on the fixed code (+2.23 on the old), on the same 30 worlds. Buttons that were never pressed where they would work (the wood pickaxe, the furnace, placing stone) get pressed once a useless press can no longer teach the agent that the button is bad. The mask reads the game's rules, so it is a separate arm from the published, knowledge-free agents.

1. The manager's lead is not established

v1.2 scored 9.97 on the old code but 8.77 after the RSSM-2 fixes, and its lead over B2 on the fixed code is +0.50 ± 0.57. The old manager results were single seeds whose differences (v1.1 vs v1.2 vs v1.3) are all within noise on paired worlds. Two more seeds of B2, v1.2 and v2 will say whether the manager helps.

2. v2 clearly beats vanilla DreamerV3 -- but not B2

v2 drops the hand-written goals, the reward potentials and the map labels, and still beats vanilla by +2.53 ± 0.44 (fixed code). It stays at the wood tier: no stone tool in any episode. B2 leads it by +1.10 ± 0.59. Curiosity, once scaled (v2-cur), changed nothing measurable.

3. The RSSM-2 fixes did not explain the results

Training no longer sees future steps, episodes no longer leak, and non-moves no longer count as a step down. The fixed runs land within noise of the old ones (v1.2 aside). The map model's contribution was real but not driven by the leak.

4. Survival is still unsolved

Lifespans stay near the random band (255-320 steps); thirst, hunger and mobs still kill most episodes. The longest-living fixed run is v1.2 (319 steps), which crafts less than before.

5. What would settle the rest

Training seeds: three per arm for B2, v1.2 and v2 (two more each are queued), and ideally for vanilla. Then a knowledge-free comparison: v2 without the rule mask needs a mask that can tell "does nothing" from "not yet learned" (B3 and B3-fix could not).